BUSINESS INTELLIGENCE LAB

EXPERIMENT 7

Integration of Advanced Analytics with BI Reports using Microsoft Power BI

Report

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1. AIM

To integrate advanced analytics with Business Intelligence (BI) reports and generate meaningful business insights, recommendations, and executive-level presentations using Microsoft Power BI.

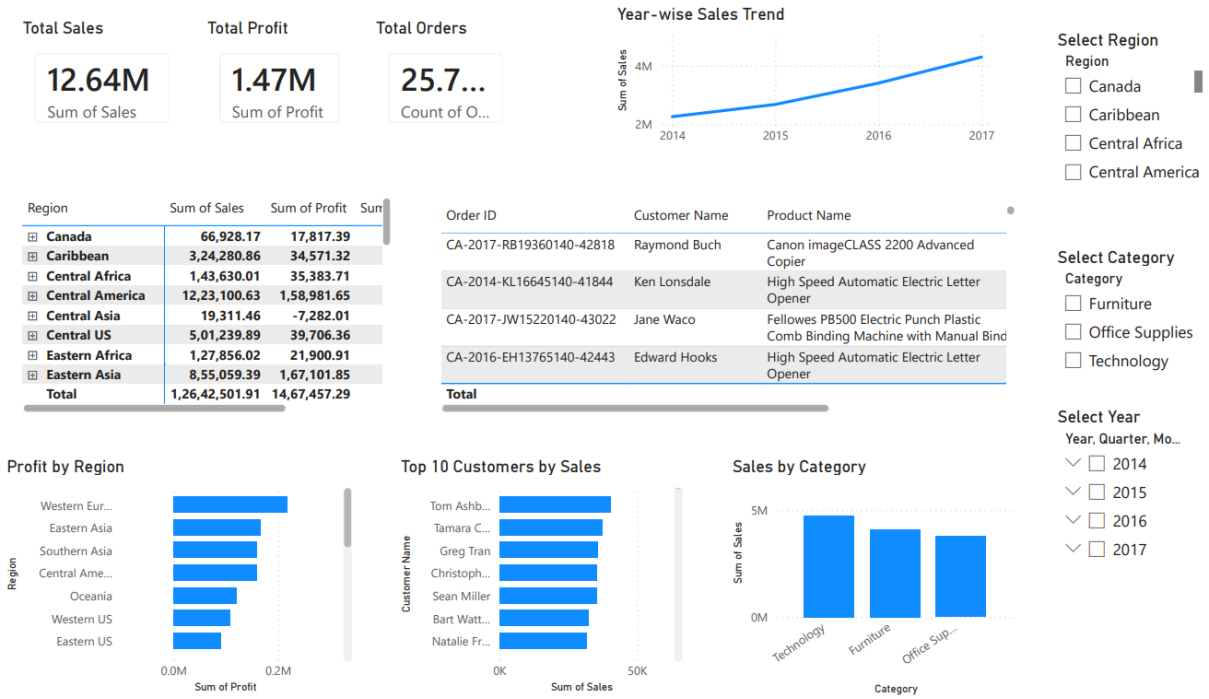
2. SOFTWARE REQUIREMENTS

1. Microsoft Power BI Desktop
2. Global Superstore Dataset
3. Existing Power BI reports from Experiment 5 and Experiment 6
4. Microsoft Word
5. Microsoft PowerPoint

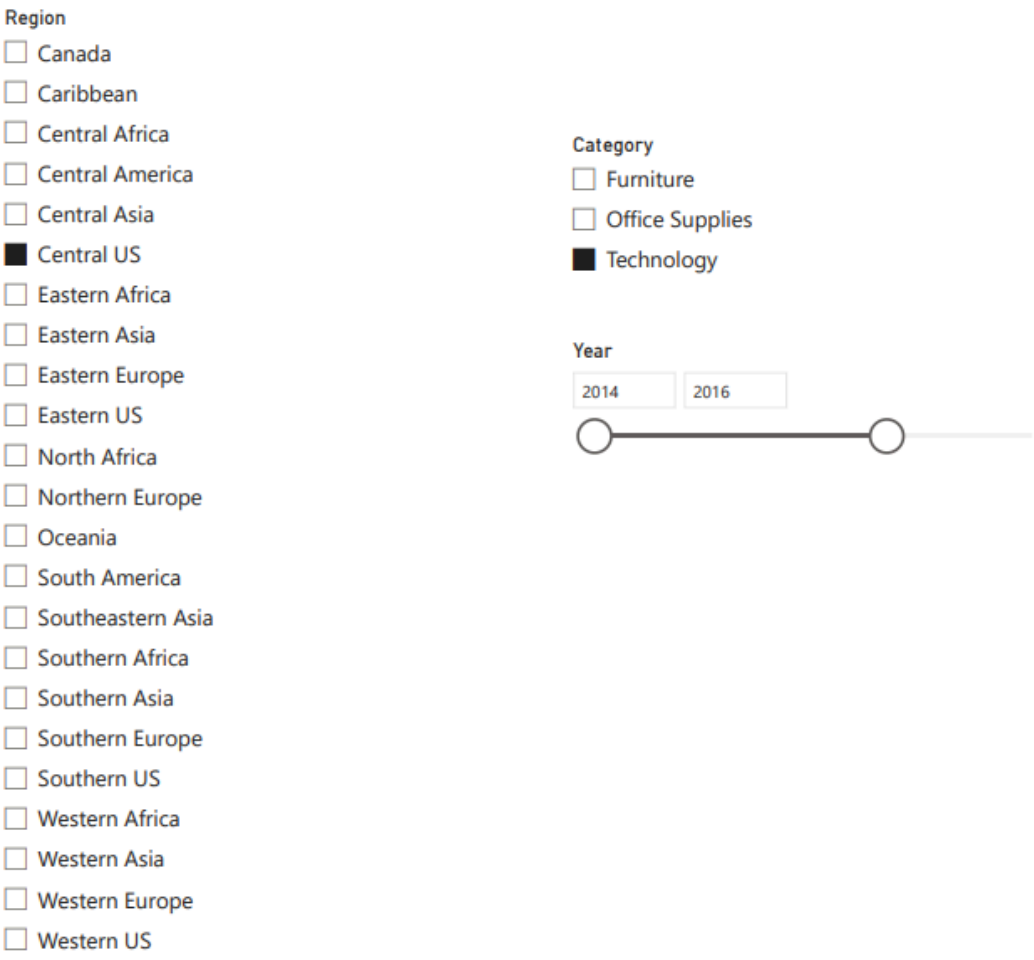
3. STEPWISE PROCEDURE

Step 1: Open Existing Power BI Reports

1. Open Microsoft Power BI Desktop.
2. Open the previously created Power BI report from Experiment 5.
3. Open the Power BI report from Experiment 6.
4. Review the existing dashboards and their visualizations.
5. Identify the important KPIs and charts available in both reports.


Total Sales
12.64M
Sum of Sales
Total Profit
1.47M
Sum of Profit
Total Orders
25.7...
Count of O...
Year-wise Sales Trend
Sum of Sales
4M
2M
2014
2015
2016
2017
Select Region
Region
Canada
Caribbean
Central Africa
Central America
Region Sum of Sales Sum of Profit Sum
Canada 66,928.17 17,817.39
Caribbean 3,24,280.86 34,571.32
Central Africa 1,43,630.01 35,383.71
Central America 12,23,100.63 1,58,981.65
Central Asia 19,311.46 -7,282.01
Central US 5,01,239.89 39,706.36
Eastern Africa 1,27,856.02 21,900.91
Eastern Asia 8,55,059.39 1,67,101.85
Total 1,26,42,501.91 14,67,457.29
Order ID Customer Name Product Name
CA-2017-RB19360140-42818 Raymond Buch Canon imageCLASS 2200 Advanced Copier
CA-2014-KL16645140-41844 Ken Lonsdale High Speed Automatic Electric Letter Opener
CA-2017-JW15220140-43022 Jane Waco Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind
CA-2016-EH13765140-42443 Edward Hooks High Speed Automatic Electric Letter Opener
Total
Select Category
Category
Furniture
Office Supplies
Technology
Select Year
Year, Quarter, Mo...
2014
2015
2016
2017
Profit by Region
Western Eur...
Eastern Asia
Southern Asia
Central Ame...
Oceania
Western US
Eastern US
Region
0.0M
0.2M
Sum of Profit
Top 10 Customers by Sales
Tom Ashb...
Tamara C...
Greg Tran
Christoph...
Sean Miller
Bart Watt...
Natalie Fr...
Customer Name
0K
50K
Sum of Sales
Sales by Category
5M
0M
Sum of Sales
Technology
Furniture
Office Sup...
Category


Total Sales
12.64M
Total Sales
Total Profit
1.47M
Total Profit
Total Profit Margin
0.12
Profit Margin (%)
Total Orders
26K
Total Orders
Sales by Region
Western ...
Central A...
Oceania
Southeas...
Southern...
Eastern ...
Western ...
Eastern US
Northern...
Region
0M
1M
2M
Sum of Sales
Monthly Sales Trend
1.5M
1.0M
0.5M
Sum of Sales
January
February
March
April
May
June
July
August
September
October
November
December
Month
Profit by Segment
0.8M
0.6M
0.4M
0.2M
0.0M
Sum of Profit
Consumer
Corporate
Home Office
Segment
Sales by Category
5M
0M
Sum of Sales
Technolo...
Furniture
Office Su...
Category
Top 10 Customers by Sales
Tom Ashbrook
Tamara Chand
Greg Tran
Christopher ...
Sean Miller
Bart Watters
Customer Name
0K
50K
Total Sales
Total Profit by Category
0.5M
0.0M
Total Profit
Technology
Office Supplies
Furniture
Category

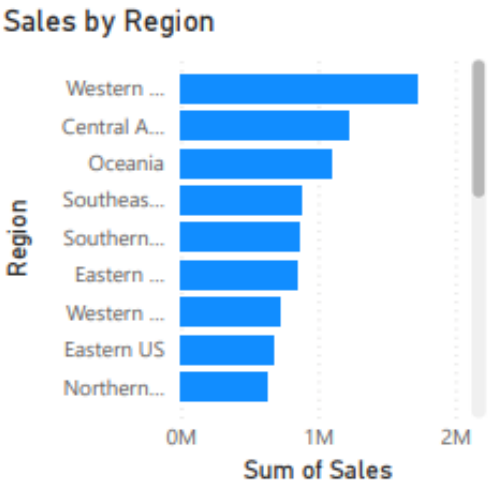


Step 2: Analyze Overall Business Performance

1. Examine the KPI cards available in the Experiment 6 dashboard.
2. Observe Total Sales, Total Profit, Total Orders, and Profit Margin.
3. Record the overall business performance.
4. Compare the sales and profit values to understand business profitability.

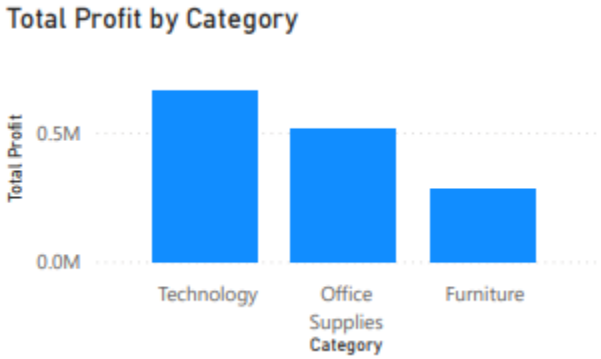
Step 3: Analyze Sales Performance by Region

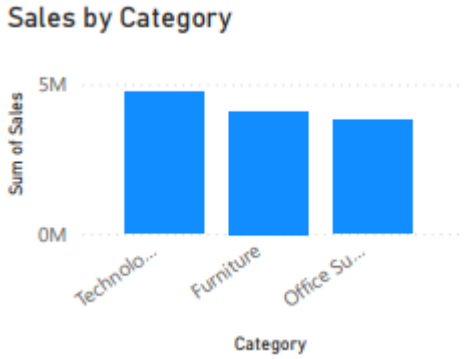
1. Open the Sales by Region visualization.
2. Compare the sales performance of different geographical regions.
3. Identify the best-performing and worst-performing regions.
4. Observe the contribution of each region to total sales.
5. Record the regional performance findings.



Step 4: Analyze Profitability by Category

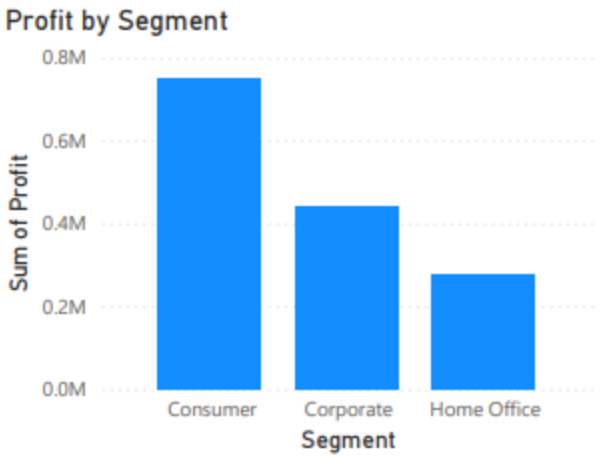
1. Open the category-wise sales and profit visualization.
2. Compare the performance of Furniture, Office Supplies, and Technology.
3. Identify the most profitable and least profitable categories.
4. Observe the relationship between sales and profit.
5. Record the category-wise findings.





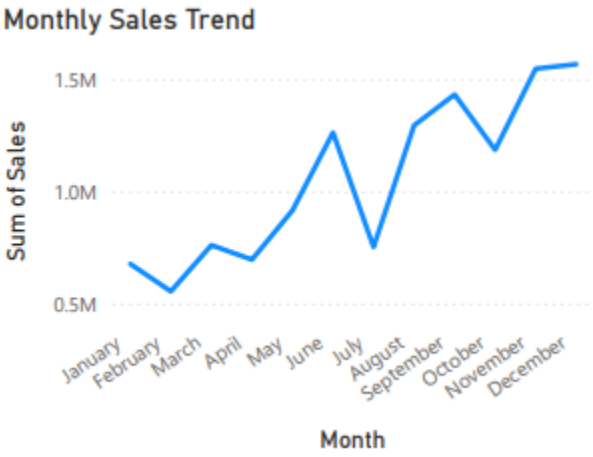
Step 5: Analyze Customer Segment Performance

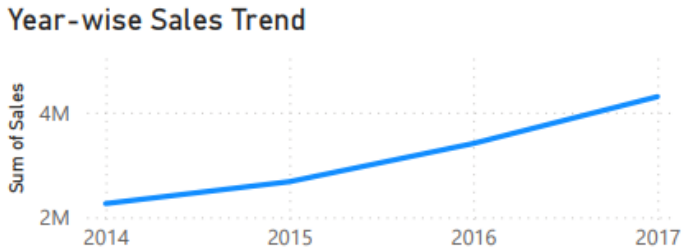
1. Open the Profit by Segment visualization.
2. Compare Consumer, Corporate, and Home Office customer segments.
3. Analyze their contribution to overall sales and profit.
4. Identify the customer segments performing well.
5. Record the observations.



Step 6: Analyze Monthly and Yearly Sales Trends

1. Open the Monthly Sales Trend visualization in Experiment 6.
2. Observe the changes in sales over different months.
3. Open the Year-wise Sales Trend visualization from Experiment 5.
4. Compare sales performance across different years.
5. Identify periods of increasing or decreasing sales.





Step 7: Identify Business Strengths

1. Review all the existing dashboard visualizations.
2. Identify the areas where the business is performing well.
3. Note at least two major business strengths.
4. Support the findings using relevant KPIs and charts.

Step 8: Identify Business Weaknesses

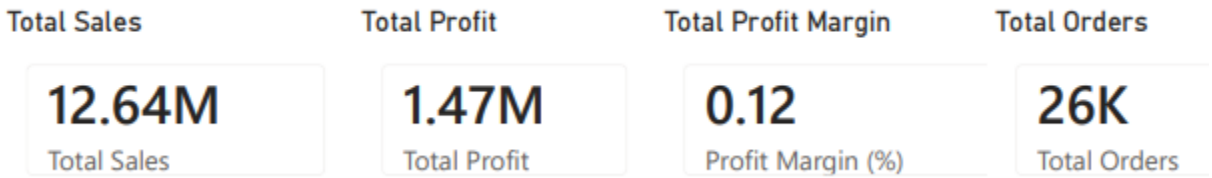
1. Examine the low-performing regions and product categories.
2. Analyze the areas with comparatively low profitability.
3. Identify at least two major business weaknesses.
4. Record the findings based on the dashboard analysis.

Step 9: Identify Business Opportunities

1. Analyze the performance of profitable product categories.
2. Identify regions with potential for improvement.
3. Explore opportunities for increasing customer engagement.
4. Identify at least two opportunities for business growth.

Step 10: Prepare Business Recommendations

1. Use the identified strengths, weaknesses, and opportunities.
2. Suggest strategies to improve overall profitability.
3. Recommend ways to improve the performance of weak regions and categories.
4. Suggest methods for increasing customer engagement and sales.
5. Prepare practical recommendations for management.



Step 11: Prepare the Business Insight Report

1. Open Microsoft Word and create a new document.
2. Add the experiment title, aim, and objectives.
3. Mention the dashboards used for analysis.

4. Include the key business findings and SWOT analysis.
5. Add suitable business recommendations.
6. Insert relevant screenshots of the existing Power BI dashboards.
7. Write the conclusion and result.

4. OBSERVATIONS

The following observations were obtained from the existing Power BI dashboards:

1. Total Sales: 12.64 Million
2. Total Profit: 1.47 Million
3. Total Orders: Approximately 26K
4. Profit Margin: Approximately 12%
5. Best Performing Region: Western Europe
6. Worst Performing Region: Central Asia
7. Most Profitable Category: Technology
8. Least Profitable Category: Furniture

The dashboards also provided information about customer segment performance, monthly sales trends, and actual versus target performance.

5. RESULT

Successfully analyzed the existing Power BI dashboards from Experiments 5 and 6, evaluated overall business performance, identified strengths, weaknesses, and opportunities, and developed suitable business recommendations. A Business Insight Report and Executive Presentation were prepared to support effective business decision-making.